F78AA1 – December 2012 – Solutions.

1.

i	v	d	$i^{(3)}$	$d^{(4)}$
0.075	0.930233	0.069767	0.073199	~~0.071456~~ 0.071671

[1 mark each, but ½ mark lost from each if < 6 d.p.]

[Total 4]

2.(a) By definition $PV(s) = \sum_{r=1}^{n} c_r v(s,t_r)$ [1]

$= \sum_{r=1}^{n} c_r v(s,t_r) \frac{v(t_r,t)}{v(t_r,t)}$ [1]

$= \sum_{r=1}^{n} c_r v(s,t) A(t_r,t)$ [1]

$= \left[\sum_{r=1}^{n} c_r A(t_r,t)\right] v(s,t)$ [1]

$= ACC(t)\, v(s,t)$

(b) (i) For convenience define 1/1/13 to be $t=0$ and use time unit 1 year.

$PV(0) = \sum_{r=1}^{5} c_r v(0,t_r)$ [1]

$= 500\, v(0,0)$
$+300\, v(0,1)$
$-100\, v(0,2)$ [1]
$-200\, v(0,3)$
$-300\, v(0,4)$

$$= 500 + 300(1.02)^{-1} + 100(1.02)^{-2} - 200(1.02)^{-2}(1.005)^{-4} - 300(1.02)^{-2}(1.005)^{-8}$$ [1]

$$= 500 + 300 \times 0.980392 - 100 \times 0.961169 - 200 \times 0.942183 - 300 \times 0.923573$$

$$= 232.4922 \text{ or } £232.49 \text{ rounded}$$ [1]

(b) $Acc(5) = PV(0) \times A(0,5)$ [1]

$= 232.4922 \times (1.02)^{2}(1.005)^{12}$

$= 256.80$ [1]

[Total 10]

3(a) $(\bar{I}s)^{i}_{\overline{n}|} = (1+i)^{n-1} + 2(1+i)^{n-2} + \ldots + n(1+i)^{0}$ [1]

$\Rightarrow (1+i)(\bar{I}s)^{i}_{\overline{n}|} = (1+i)^{n} + 2(1+i)^{n-1} + \ldots + n(1+i)^{1}$ [1]

$\Rightarrow (1+i)(\bar{I}s)^{i}_{\overline{n}|} - (\bar{I}s)^{i}_{\overline{n}|} = (1+i)^{n} + (1+i)^{n-1} + \ldots + (1+i)^{1} - n$ [1]

$\Rightarrow i(\bar{I}s)^{i}_{\overline{n}|} = \ddot{s}^{i}_{\overline{n}|} - n$

$\Rightarrow (\bar{I}s)^{i}_{\overline{n}|} = \dfrac{\ddot{s}^{i}_{\overline{n}|} - n}{i}$ [1]

(b) $PV = 20000\, a^{0.03}_{\overline{10}|}$ [1]

$+ (1.03)^{-10}\left[22000\, a^{0.03}_{\overline{5}|} - 2000\,(\bar{I}a)^{0.03}_{\overline{5}|}\right]$ [1]

$+ (1.03)^{-10}(1.03)^{-5}\left[12000\, a^{0.04}_{\overline{5}|} - 2000\,(\bar{I}a)^{0.04}_{\overline{5}|}\right]$ [1]

$a^{0.03}_{\overline{10}|} = 8.530203$

$a^{0.03}_{\overline{5}|} = 4.579707$

$(\bar{I}a)^{0.03}_{\overline{5}|} = \dfrac{\ddot{a}_{\overline{5}|} - 5v^{5}}{0.03} = 13.46848$ [2]

$a^{0.04}_{\overline{5}|} = 4.451822$

$(\bar{I}a)^{0.04}_{\overline{5}|} = 13.006484$

Hence PV = ~~[illegible]~~ 243,123.25 [2]

[Total 10]

(4) (a) (i) Repayment $X = \frac{300,000}{a_{\overline{27}|}^{0.06}} = £23,468.02$ [1]

(ii) Cap. o/s $= X a_{\overline{15}|}^{0.06} = £227,927.25$ [1]

(iii) Int. cont. $=$ Cap o/s $\times 0.06 = £13,675.64$ [1]

(6) After two years, the cap o/s is

$227,927.25 \times (1.09)^2 = £270,800.37.$ [1]

Let the new level repayment be X'.

Year 13. Repayment $= X'$

Int. cont. $= 270,800.37 \times 0.09 = 24,372.03$

Cap. cont. $= X' - 24,372.03$

Cap. o/s $= 295,172.40 - X'$

Year 14 Repayment $= X'$

Int. cont. $= 26,565.52 - 0.09X'$

Cap. cont $= 1.09X' - 26,565.52$

Cap. o/s $= 321,737.92 - 2.09X'$ [3]

[or directly: $227927.25(1.09)^4 - X' s_{\overline{2}|}^{0.09}$]

Hence $X' a_{\overline{13}|}^{0.06} = 321,737.92 - 2.09X'$ [1]

$\Rightarrow X' = \frac{321,737.92}{a_{\overline{13}|}^{0.06} + 2.09}$ [1]

$=$ ~~48,916.52~~ 29,402.11 [1]

[Total 10]

5 (a) Loan

Find the breakeven point.

$$55\, a_{\overline{7}|}^{0.08} = 286.35$$
$$55\, a_{\overline{8}|}^{0.08} = 316.065 \qquad [2]$$

Hence breakeven point is $t = 8$, and the net profit then is:

$$16.065\,(1.08)^8 = 29.735 \qquad [1]$$

Hence accumulated profits are

$$29.735\,(1.04)^2 + 55(1.04) + 55 + 80 \qquad [1]$$

$$= 224.36 \qquad [1]$$

Bond

Net profit each year $= 55 - 0.05 \times 300 = 40$ [1]

Accumulated net profit $= 40\, s_{\overline{7}|}^{0.04} = 480.24$ [1]

Redemption proceeds $= 1.10 \times 300 = 330$ [1]

Sale proceeds $= 80$ [1]

Total accumulated profit = 230.24 [1]

Hence it is better to issue the bond.

(i) Net of CGT, the redemption proceeds are:

$$330 - 0.4(330 - 300) = 318. \qquad [1]$$

Set up an equation of value at $t = 10$ (yrs)

$$f(i) = 300(1+i)^{10} - 11.25\, s^{i}_{\overline{10}|} - 318 \qquad [1]$$

$$\uparrow$$

$$0.75 \times 0.05 \times 300 \qquad [1]$$

and find i^* such that $f(i^*) = 0$

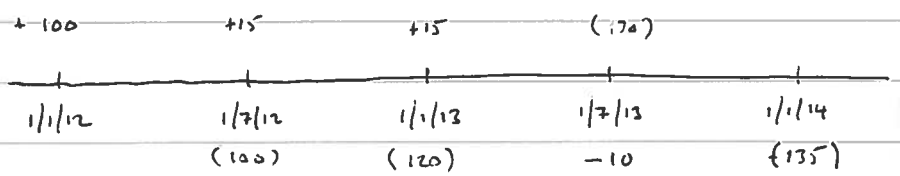
$$f(0.04) = 444.07 - 135.07 - 318 = -9.00$$
$$f(0.05) = 488.67 - 141.50 - 318 = +29.17 \qquad [2]$$

Hence $i^* \cong 0.04 + 0.01 \dfrac{9.00}{9.00 + 29.17}$

$$= 0.0424. \qquad [1].$$

[Total 16].

6. (a)



Let i_1 be the effective yield per half year in 2012.
The equation of value is

$$f(i_1) = 100(1+i_1^*)^2 + 15(1+i_1) - 120 \qquad [1]$$

Hence $1+i_1^* = \dfrac{-15 \pm \sqrt{15^2 + 400\times 120}}{200}$ [1]

$$\Rightarrow 1+i_1^* = 1.0230$$

Let i_2 be the effective rate of interest per half year in 2013. The equation of value is

$$f(i_2) = 135(1+i_2)^2 - 10(1+i_1) - 135 \qquad [1]$$

Hence $1+i_2^* = \dfrac{10 \pm \sqrt{10^2 + 4\times 135^2}}{270}$ [1]

$$\Rightarrow 1+i_2^* = 1.0377$$

Hence the LIRR per annum is

$$\left[(1.023)^2(1.0377)^2\right]^{1/2} - 1 \qquad [1]$$

$$= 0.0616 \qquad [1]$$

(b) $TWRR = \left[\frac{100}{100}\,\frac{120}{115}\,\frac{130}{135}\,\frac{135}{120}\right]^{1/4} - 1$ [1]

$= 0.0632$ [1]

[Total 8]